HDOC DAY 58

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, arr[], and an integer k such that 0 < k ≤ size. Pass the array to a function named kth_smallest() using its address. Use pointers to pass the array between the calling function and kth_smallest(). The function kth_smallest() must:

- Arrange the elements of arr[] in ascending order using bubble sort algorithm.
- Find the element that occupies the kth position in the sorted array.
- Display the kth smallest element.
- Consider duplicate elements as separate elements.
- Display -1 if k is invalid.

CODE

```
#include <stdio.h>


void kth_smallest(int *arr, int size, int k)
{
   int i, j, temp;


   if (k <= 0 || k > size)
   {
      printf("-1");
      return;
   }


   for (i = 0; i < size - 1; i++)
```

```
    {
        for (j = 0; j < size - i - 1; j++)
        {
            if (*(arr + j) > *(arr + j + 1))
            {
                temp = *(arr + j);
                *(arr + j) = *(arr + j + 1);
                *(arr + j + 1) = temp;
            }
        }
    }

    printf("%d", *(arr + k - 1));
}

int main()
{
    int size, k, i;

    scanf("%d", &size);

    if (size <= 0)
        return 0;

    int arr[size];

    for (i = 0; i < size; i++)
        scanf("%d", arr + i);
```

```
    scanf("%d", &k);

    kth_smallest(arr, size, k);

    return 0;
}
```